

Position-Based T-S Fuzzy Power Management for Tram With Energy Storage System

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Abstract—Energy storage systems (ESSs) play a significant role in performance improvement of future electric traction systems. This paper investigates an ESS based on supercapacitors for trams as a reliable technical solution with considerable energy saving potential. Operating the ESS onboard a tram brings the following benefits: reduction of peak power demands, decrease of power consumption, improvement of performance, and efficiency. Whereas the commonly used ESS power management (PM) has been dealing only with PM without vehicle position knowledge, the main emphasis of the presented research is put on position-based Takagi–Sugeno fuzzy (T-S fuzzy) PM for human-driven trams with an ESS. The tram position, located by, e.g., a Global Positioning System module, is used to predict future energy demands and power flows. This information is used in the preparation of the supercapacitor energy level and its power flow control. The optimal ESS power flow with respect to the tram position is a very complex problem. Therefore, soft computing techniques are used in this paper. The main part of the proposed ESS PM is the T-S fuzzy controller. The T-S fuzzy reduces problem dimensionality using expert knowledge represented by the rules. The output membership functions were optimized by differential evolution. The designed ESS PM was verified on real traffic data.

Index Terms—Differential evolution (DE), energy storage, fuzzy control, power management (PM), power system modeling.

I. INTRODUCTION

THIS RESEARCH has been motivated by industrial demand for a design of a new generation of energy storage systems (ESSs) based on supercapacitors for light rail vehicles. The accumulation of electrical energy presents a challenge that has been being responded to by numerous specialists and scientists. ESSs as a necessary energy-buffering element are becoming more and more important. Storage systems play a crucial role in many areas: 1) improvement of the operating

stability of the power system, mainly in relation to renewable power sources [1]–[4]; 2) improvement of the performance and efficiency of the electric traction system using the accumulation of the vehicle braking energy [5]–[10] and power supply for short-time experiments such as the Tokamak coil power supply (e.g., flywheels and superconductive coils).

ESSs store available energy in the appropriate medium. From the vehicle system point of view, we can think of the following: batteries [6], [11], [12], supercapacitors [5], [6], [10], and flywheels [4]. Generally, the type and dimensioning of the ESS is dependent on the power and duration of the recuperation, which determine the total energy stored in the ESS. The next factor influencing the selection of an ESS is a requirement for the distance that the tram has to pass without the catenary supply (e.g., cruising in historical downtowns). Our research has been focused on the tram on the line no. 4 in Pilsen representing a typical track with short high-power braking. Moreover, this tram does not require the cruising without overhead line (OHL) supply. Therefore, the ESS based on supercapacitors is the best choice (poor power density of batteries leads to a high mass of batteries to meet the required power demand of our tram). Supercapacitors have specific advantages: 1) high power density, reaching values ranging from 7 000 to 18 000 W/kg; 2) long operating lifetime [7], good performance in high as well as low temperatures, and high efficiency; 3) high speed and high efficiency of charge and discharge [13]; 4) minimal maintenance requirements.

The selected power management (PM) plays a crucial part in the efficiency of the ESS [14]–[16]. PM controls energy distribution and power flow. Commonly used methods such as dynamic programming [17], polynomial control strategy [18], [19], neural networks [20], fuzzy systems, and machine learning techniques [15] are mainly used only for PM without vehicle position knowledge.

On the contrary, this paper deals with a PM strategy using the position [e.g., Global Positioning System (GPS)] for a tram with an ESS. The measured tram consumed and recuperated power levels, which depend on the tram location, are saved to PM lookup tables (LUTs). The Takagi–Sugeno fuzzy (T-S fuzzy) [21] PM controller uses prior expert knowledge represented by rules and the information from energy LUTs to control the actual power of the ESS. Output membership functions of the T-S fuzzy were optimized by a heuristic optimization technique known as differential evolution (DE) [22], [23]. DE is a derivative-free function optimization over continuous space.

This paper is organized as follows: Section II describes a tram with an ESS based on supercapacitors, Section III presents basic PM strategies, Section IV describes the position-based

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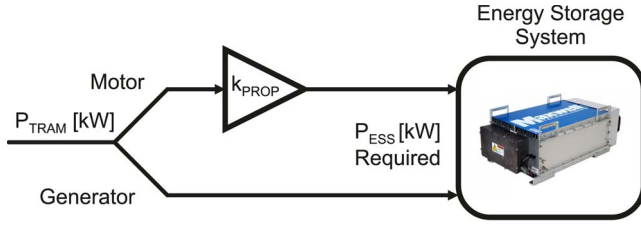


Fig. 3. Proportional PM.

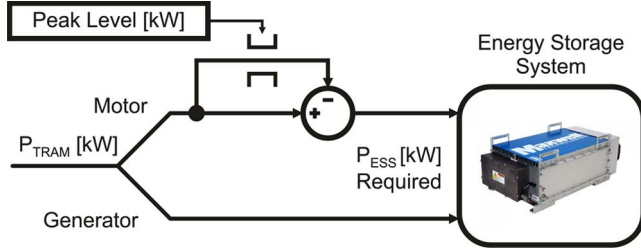


Fig. 4. Peak shaving PM.

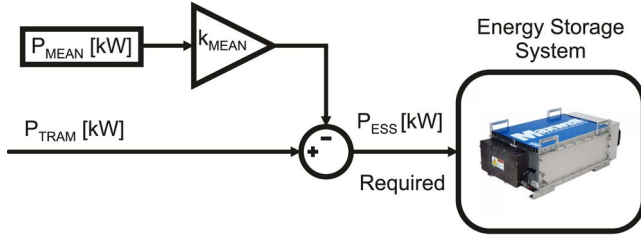


Fig. 5. PM based on mean power.

tram braking is fully accumulated into the energy storage. This principle is outlined in Fig. 3.

B. Peak Shaving PM

Peak shaving PM uses an ESS to shave the power peaks. If the power demand is higher than the constant peak level, then the peak is shaved. The maximal consumed power from the OHL is matched to the *peak level* parameter. The difference between the tram power demand and this constant is covered from the ESS. It is shown in Fig. 4. The recuperated energy is fully accumulated into the energy storage.

C. PM Based on Mean Power

Mean power PM is based on the knowledge of the tram line consumption profile (the whole tram line—between the terminals). The value of the mean consumed power is calculated from this profile, for each direction separately. The mean power P_{MEAN} is multiplied by the constant $k_{\text{MEAN}} (0 \div 1)$. If k_{MEAN} is equal to 1 and the energy storage capacity is sufficient, the power consumed from the OHL will be equal to the constant P_{MEAN} . This strategy is shown in Fig. 5.

IV. T-S FUZZY PM

PM based on the T-S fuzzy controller [21] uses the previously measured tram line energy consumption profile depending on the tram position. These energy consumption profiles are spe-

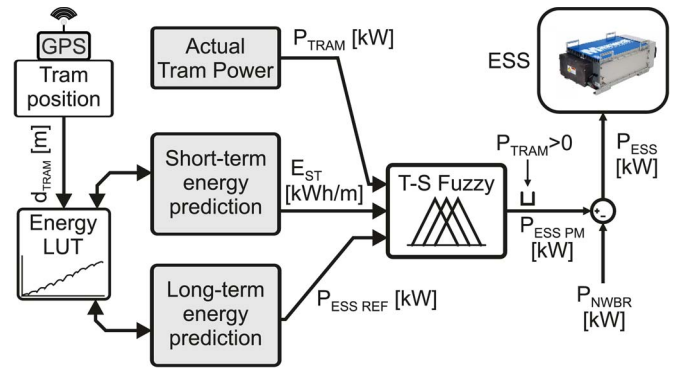


Fig. 6. T-S fuzzy PM based on GPS location.

cific for each tram line and could be adaptively updated and shared between trams on the same line. The tram energy is computed as a discrete time integral of the actual tram power and is saved into LUTs corresponding to the tram position located by, e.g., GPS. The tram energy from the previous measurements saved in the LUTs is used for long- and short-term prediction. The T-S fuzzy PM controls—via implemented rules and their outputs—the actual supercapacitor power flow depending on the actual tram power demands and short- and long-term energy requirements (see Fig. 6).

The T-S fuzzy controller consist of three inputs: actual tram power P_{TRAM} and two tram position forecasted values: short-term energy prediction E_{ST} and long-term energy prediction represented by the ESS power reference $P_{\text{ESS REF}}$. The maximal positive output power of the T-S fuzzy is limited by the actual tram power demand. The regenerative power, which is not possible to recuperate, is always accumulated in the ESS (the last term of subtracting P_{NWBR}), until the SC is fully charged or its current is reduced to prevent SC overheating.

A. Short-Term Energy Prediction

Short-term prediction is computed as the difference between the actual tram position energy and 60-m ahead energy. This energy is normalized by the length of the horizon prediction (i.e., by 60). The value of 60 m was selected as a common accelerating or braking tram distance, i.e.,

$$E_{\text{ST}} = (E_{\text{LUT}_{d_{\text{TRAM}}+60\text{m}}} - E_{\text{LUT}_{d_{\text{tram}}}}) / 60.$$

B. Long-Term Energy Prediction

In long-term prediction, the energy LUT is analyzed to forecast energy peaks and their distance from 60 m ahead to 500 m ahead. This prediction is used to avoid ESS capacity limitation and to minimize probability of wasting energy in the braking resistor. The basic principle of the peak detector is shown in Fig. 7.

The peak detector algorithm search in forecasted energies saved in the energy LUT. The energy peak is identified, while the absolute value of difference $|E_{\text{TRAM}(n)} - E_{\text{TRAM}(n-1)}|$ is higher than the selected limit. The forecasted energy peaks are used as a reference for the proportional controller of the ESS energy (see Fig. 8). The reference value for the proportional

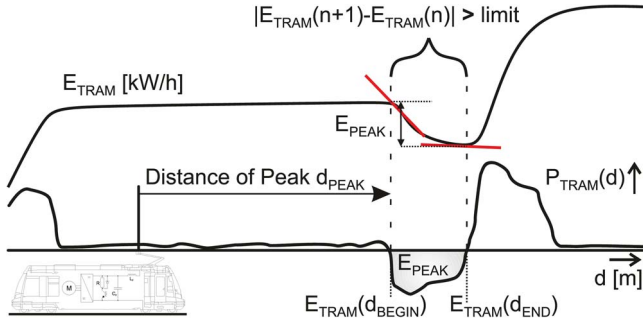


Fig. 7. Principle of long-term prediction with energy peak localization (simplified for constant speed).

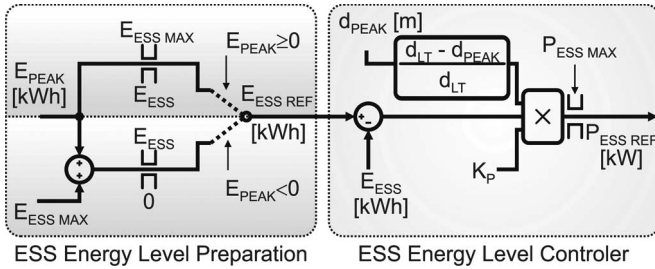


Fig. 8. ESS energy level preparation block with the proportional ESS level controller.

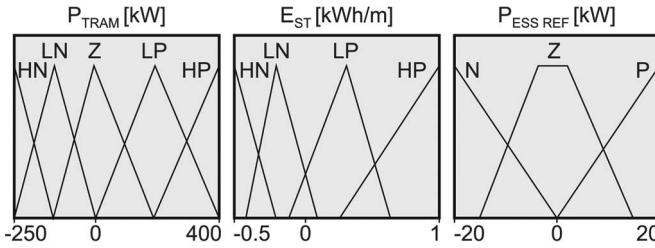


Fig. 9. T-S fuzzy input membership functions: $P_{\text{ESS REF}}$ is long-term prediction input, P_{TRAM} is actual power of the ESS tram, and E_{ST} is short-term prediction input.

regulator is dependent on the energy peak polarity, and it is limited by the max/min energy of the SC. $E_{\text{ESS REF}}$ is equal to E_{ESS} until E_{ESS} is able to cover E_{PEAK} (positive/negative). It reduces ESS energy fluctuation and losses. The difference (error) between the reference value $E_{\text{ESS REF}}$ and the actual E_{ESS} value is multiplied by the proportional gain K_P and weighted by the normalized distance value of the peak ($0 \div 1$).

C. T-S Fuzzy Controller

The T-S fuzzy controller consists of three input variables with triangular input membership functions (see Fig. 9) and 11 singleton output membership functions. The number of output membership functions is a compromise between system complexity and fitting ability.

The output level z_i of each rule is weighted by the firing strength w_i . The rule has the following form:

$$w_i = F(P_{\text{ESS REF}}) \text{ AND } F(P_{\text{TRAM}}) \text{ AND } F(E_{\text{ST}}).$$

F belongs to the membership of each input function, and AND is realized as the minimum value. The output of the T-S

$P_{\text{TRAM}} \rightarrow$		HN	LN	Z	LP	HP
$E_{\text{ST}} \uparrow$	HP	-150	-150	N -50 Z 0 P 0	20	100
	LP	-150	-100	N 0 Z 0 P 0	40	120
	LN	-100	-100	N 0 Z 0 P 20	60	140
	HN	-100	-50	N 0 Z 0 P 40	80	140
				$P_{\text{ESS REF}} \uparrow$		

Fig. 10. T-S fuzzy rules, where each cell is one rule with the corresponding partial output of the T-S fuzzy PM in kilowatts. The rules implement expert knowledge to the T-S fuzzy PM.

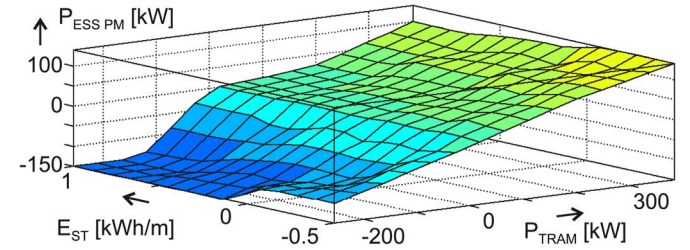


Fig. 11. Transfer function of manually set T-S fuzzy controller for $P_{\text{ESS REF}} = 0$ (before optimization by DE).

fuzzy—the required ESS power $P_{\text{ESS PM}}$ —is computed as the weighted average of the rule output z_i . Thus,

$$P_{\text{ESS PM}} = \frac{\sum_{i=1}^{11} (w_i z_i)}{\sum_{i=1}^{11} w_i}.$$

D. T-S Fuzzy Rules

Fig. 10 presents a T-S fuzzy rule, where each cell represents one AND rule with the corresponding inputs and rule outputs z_i (in the middle of each cell in kilowatts). The long-term prediction (variable $P_{\text{ESS REF}}$) affects the actual ESS power only when the tram power P_{TRAM} is near zero. This property reduces OHL and SC resistance losses. An example of the rule for the bottom-right corner follows.

When the short-term energy prediction E_{ST} is high negative (HN) and actual power P_{TRAM} is high positive (HP), then the output z_i , which represents ESS power, is equal to 140 kW. It also means that if the tram actually consumes high power and rapid braking is forecasted in the 60-m ahead distance, then the power of the ESS is high, i.e., 140 kW.

Fig. 11 presents the resulted transfer function for the intuitively (manually) set T-S fuzzy controller (shown for $P_{\text{ESS REF}} = 0$). The fuzzy transfer function approximates outputs of rules presented in Fig. 10.

V. DE-BASED T-S FUZZY OPTIMIZATION

The DE [22], [23] is a multipopulation heuristic technique for global function optimization over continuous space. The T-S fuzzy PM is preoptimized offline [26] using evolution of 11

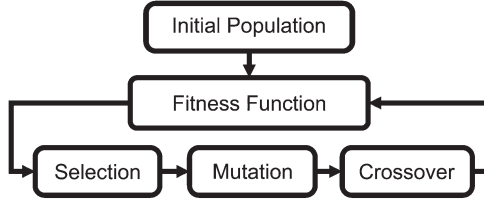


Fig. 12. DE basic scheme.

partial rule outputs $z_{(i)}$. The evaluation function (called fitness function) is consumed energy of the ESS tram drive simulation.

At the start (see Fig. 12), the initial population (set of partial outputs $z_{(i)}$) of the T-S fuzzy rule output is randomly initialized between ± 150 kW. These values correspond with the expected limits of the ESS power $P_{ESS PM}$ (determined by the number of supercapacitors, their maximum power, etc.). The initial population matrix has 30 members with 11 variables (rule outputs). The PM with the randomly selected output rules is verified by the simulation of the tram line.

The fitness function is computed as a total sum of the ESS tram consumed energy (with respected OHL losses, etc.).

The better members from the population have lower values. The goal of DE is to minimize the fitness function in the future steps. The best 30 population members and their fitness are saved in the selection block. Members are sorted accordingly to their fitness. During the mutation process, the population is randomly reordered twice to create populations Z_{REORD_1} and Z_{REORD_2} . The difference between these two populations is multiplied by the constant $C_{MUT} = 0.2$, and the result is added to the original population.

This new mutated population is mixed with the original population in the crossover block. The member variables (rule outputs) of the original and mutated population are swapped with each other with defined probability $P_{PROB} = 0.2$. In the selection block, this population is compared with the old population, and the better 30 members survive to the next iteration.

The number of DE iterations was selected as 90. The 90 iterations lead to a similar best value convergence from different initial conditions (different initial population). The next iterations bring very slow improvement, and overfitting phenomena can appear.

Mutation

$$Z_{MUT} = Z + C_{MUT}(Z_{REORD_1} - Z_{REORD_2}).$$

The resulted transfer function for the best optimized T-S fuzzy controller for our preselected rules is shown in Fig. 13. The output ESS power ($P_{ESS PM}$) of the T-S fuzzy PM is lower than manually selected rule outputs (see Fig. 11). This leads to lower ESS joule losses than the original T-S fuzzy PM.

Fig. 14 presents the behavior of the T-S fuzzy PM. It is evident that the T-S fuzzy PM balances OHL losses and ESS losses via the proper ESS power (P_{ESS}) control. The short-term prediction causes smooth releasing of the stored SC energy (E_{ESS}). The effect of the long-term prediction is noticeable from the E_{ESS} waveform. The ESS tries to prepare for the future demands represented by the reference value $E_{ESS REF}$.

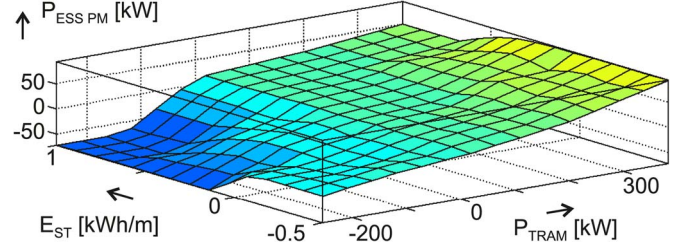
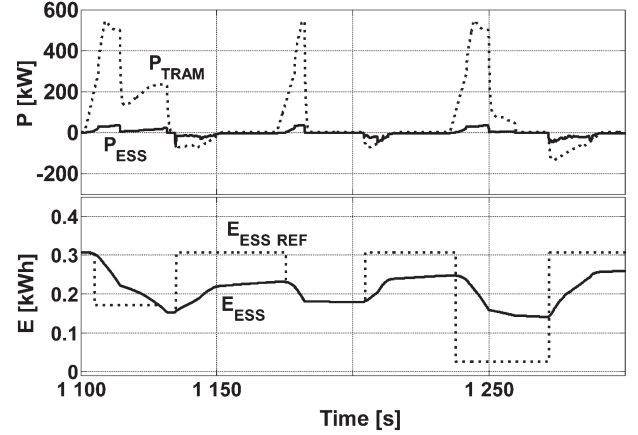
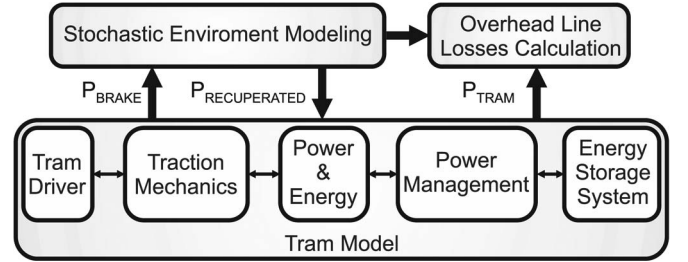
Fig. 13. Transfer function of the T-S fuzzy controller optimized by DE $P_{ESS REF} = 0$.Fig. 14. Simulation of the T-S fuzzy PM after optimization. P_{TRAM} is the total tram power, P_{ESS} is the ESS power controlled by the fuzzy PM, and E_{ESS} is the utilizable SC energy level (SC is discharged max to $1/2 U_{SC NOMINAL}$).

Fig. 15. Basic principle of the stochastic modeling of the tram with the ESS.

(obtained by the energy LUTs) during the low P_{TRAM} power. It is visible in the SC charging before positive P_{TRAM} peaks.

VI. STOCHASTIC SIMULATION MODEL

The stochastic simulation model is used for the ESS tram environment simulation [27]. The main goal is to simulate the real tram line operated by other trams. The efficiency of the ESS tram strongly depends on a recuperative ratio and OHL losses.

The ESS tram model is composed of five mutually interacting blocks (see Fig. 15). The tram driver affects the tram mechanics (e.g., tram position, speed, etc.), and vice versa. The simulated tram is controlled by a model of the driver, and the tram operates on a real tram line profile with simulated mechanical losses, traffic, etc. The most important block from the PM point

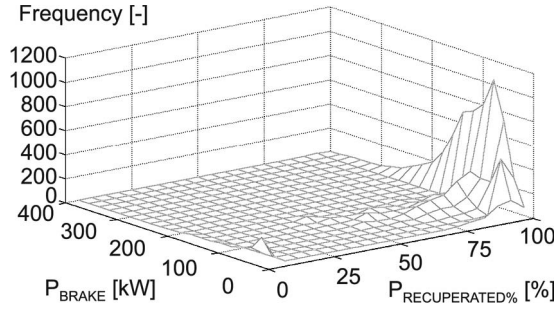


Fig. 16. Statistical frequency of the recuperative ratio in the supply sector no. 8. This histogram is used for the ESS tram recuperated ratio determination depending on P_{BRAKE} .

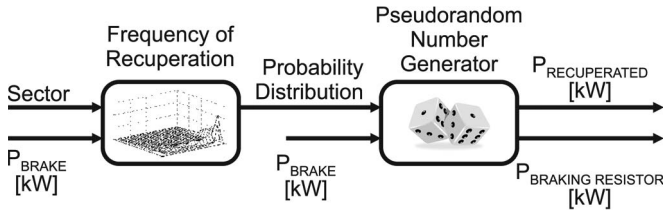


Fig. 17. Stochastic modeling of recuperative ratio.

of view is Power&Energy (see Fig. 15). This block is directly affected by the traction mechanics, and it is controlled by the PM. Other trams, which affect the recuperation and the OHL losses of the ESS tram, are stochastically simulated using the measured data.

A. Recuperated Power Calculation

During braking, the power P_{BRAKE} is divided into recuperated power $P_{\text{RECUPERATED}}$ and wasted power $P_{\text{BRAKING RESISTOR}}$. The histogram in Fig. 16 creates probability distribution for the random number generator (see Fig. 17).

Inputs of histogram selection are P_{BRAKE} and Sector. The Sector represents the part of the OHL with an independent power supply. Tram line no. 4 consists of nine sectors. Each sector has its own probability distribution function of the recuperated power ratio. The histogram in Fig. 16 shows statistical frequency in sector no. 8. All histograms are gained from measured data ($P_{\text{TRAM}} < 0$) and the $P_{\text{RECUPERATED}}$ data in each sector. The statistical frequency of the recuperation ratio (0%–100%) is computed for all tram braking power quantized by 20 kW (see Fig. 16).

The percentage of the recuperated power $P_{\text{RECUPERATED}}$ is randomly generated for selected braking power (reduced by P_{ESS} for the tram with ESS) and actual tram power sector (see Fig. 17).

B. OHL Losses Calculation

It has been mentioned before that the tram line is divided into nine independently powered sectors. These sectors are further divided into 30-m segments. The stochastic environment is represented by a stochastic load. The OHL losses calculation can be divided into two cases.

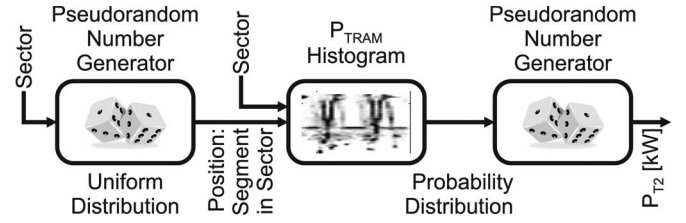


Fig. 18. Definition of the stochastic load power (ESS tram accelerating).

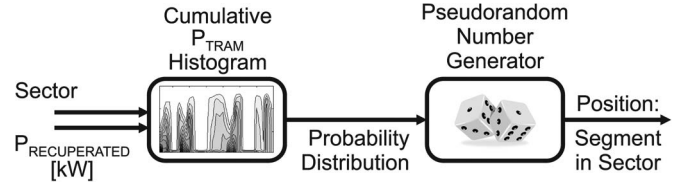


Fig. 19. Stochastic load tram distance during ESS tram recuperation.

1) **ESS Tram Accelerating:** The uniform random number generator generates the stochastic load position, and the load power is randomly generated by using a histogram obtained from measured tram line data. While the stochastic load “recuperates,” the ESS tram can take a part of the power. On the other hand, during the ESS tram acceleration, the stochastic load, which consumes the power, can enlarge the OHL losses of the ESS tram. The principle of modeling is visible in Fig. 18. The ESS tram sector and the randomly generated position (segment) determine the P_{TRAM} histogram selection. This histogram is used as a probabilistic distribution function of the stochastic load power P_{T2} (the P_{TRAM} histogram derivation is in Fig. 20).

2) **ESS Tram Braking:** If the stochastic load consumes the power, the ESS tram can recuperate the energy to the stochastic load position. The value of the recuperated power is gained from the random number generator during ESS tram braking (mentioned in Section VI-A).

The value of the recuperated ESS tram power affects the stochastic load position. The ESS tram sector and the defined recuperated power determine the cumulative histogram selection. The stochastic load position is randomly generated by using the selected cumulative histogram (see Fig. 19). These cumulative histograms are the sum of the P_{TRAM} histograms from $+\infty$ to zero (see Fig. 20).

The stochastic model was evaluated by comparison of the measurement and simulation results of the tram without an ESS. The criteria were the total energy consumed by the simulated tram and the measured tram and their recuperated energy. The results for recuperated energy can be seen in Fig. 21.

VII. EXPERIMENTAL RESULTS

The selected PM was measured and validated on a 9-kW laboratory prototype. This prototype consists of a traction induction drive and an ESS with SC and dc/dc converter (see Fig. 22). I_{OHL} is the current of the OHL, I_{TRAM} is the tram traction converter current, I_{SC} is the SC current, I_{ESS} is the ESS current, and velocity is the tram speed. A heavy transportation SC module by Maxwell Company was used as energy storage. The SC module has nominal voltage of 48 V and capacitance of

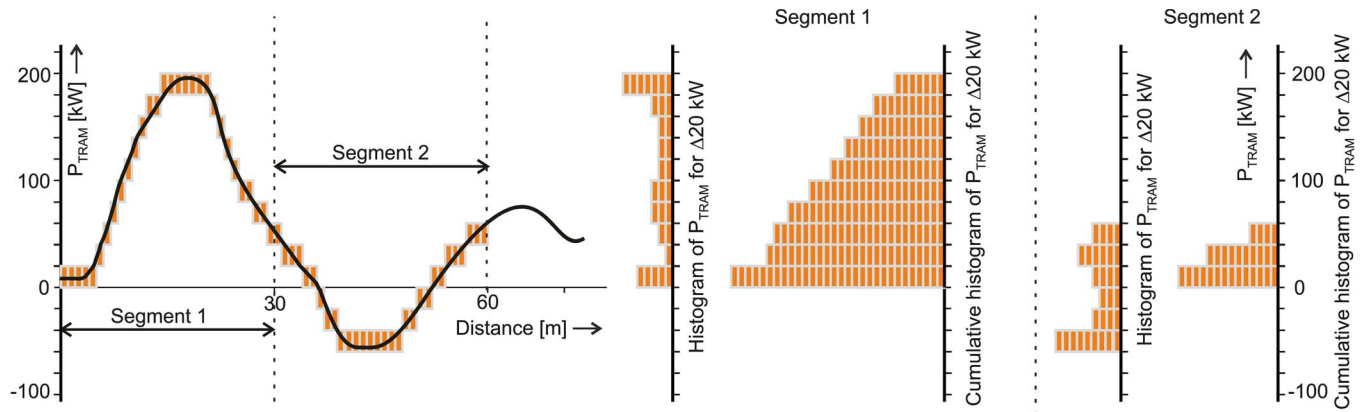


Fig. 20. Principle of histogram creation for OHL losses calculation. The histogram represents the probabilistic function of the tram power related to the tram segment. The cumulative histogram is used for determination of the stochastic load tram position during ESS tram recuperation.

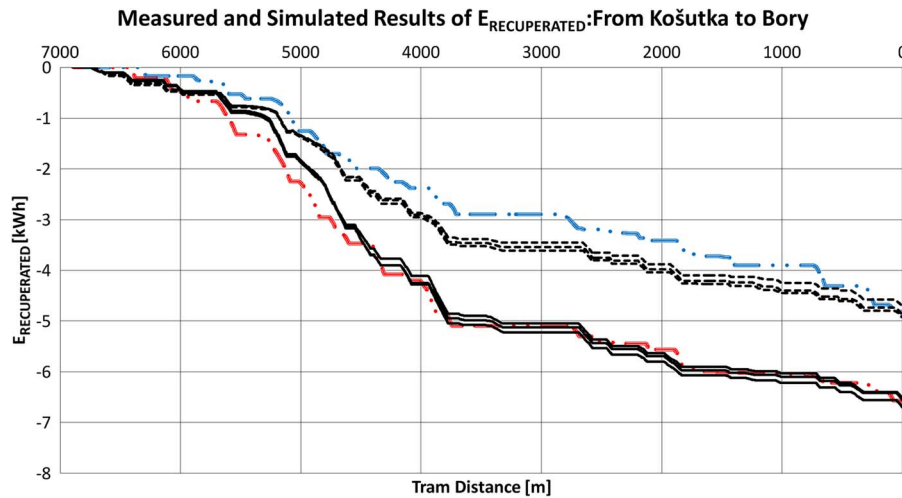


Fig. 21. Model evaluation. Red dashed-dotted line is the measured recuperated energy $E_{\text{RECUPERATED}}$, and the full black lines are the simulation results of the full load trams without an ESS. Blue dashed-dotted line is the measurement result, and dashed lines are the simulation results for the empty tram without an ESS.

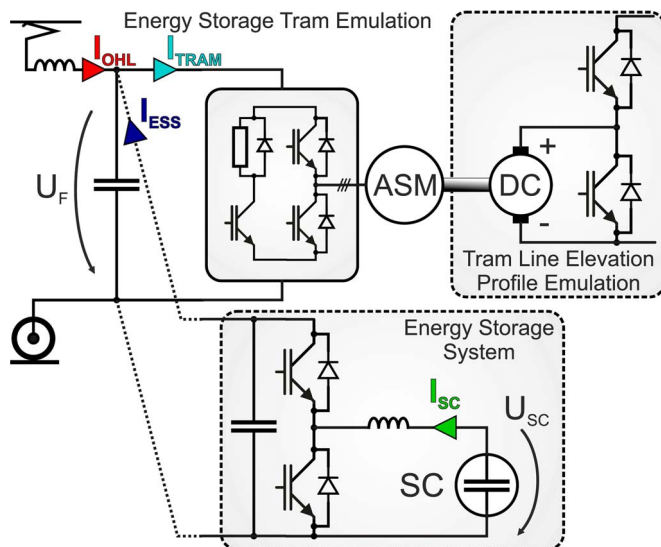


Fig. 22. Laboratory prototype of a traction induction drive with an ESS. The tram line elevation profile is emulated by a dc machine. The ESS is connected parallel to the traction drive.

83 F. The maximum current I_{SC} is limited to 40 A. The traction motor is mechanically coupled with the dc machine. It emulates the tram line elevation profile and tram moment of inertia. The OHL is emulated by a dynamo.

Measured waveforms (see Figs. 23–26) show the behavior of the mentioned PM. In case of the laboratory experiment, the shape of the I_{OHL} , I_{ESS} , and I_{TRAM} correspond to the power due to stable voltage U_F .

VIII. SIMULATION RESULTS

Simulation results of the described PM are shown in this section. The results present efficiency for different parameters of PM strategies. Each PM was tested for enabled and disabled recuperation possibility to OHL. By the enabled recuperation, a chance and an absolute value of the recuperated energy to the OHL correspond to the measured data of the real tram line in Pilsen. On the other hand, disabled recuperation means that the tram energy recovering to the OHL is forbidden (upon the customer's demand). This situation is current in many

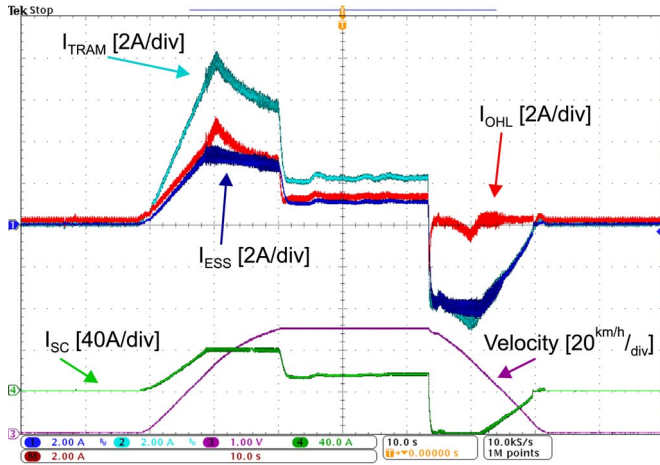


Fig. 23. Measurement results of the proportional control PM for $k_{PROP} = 0.5$: The ESS (I_{ESS}) covers I_{TRAM} by 50% until I_{SC} is not limited.

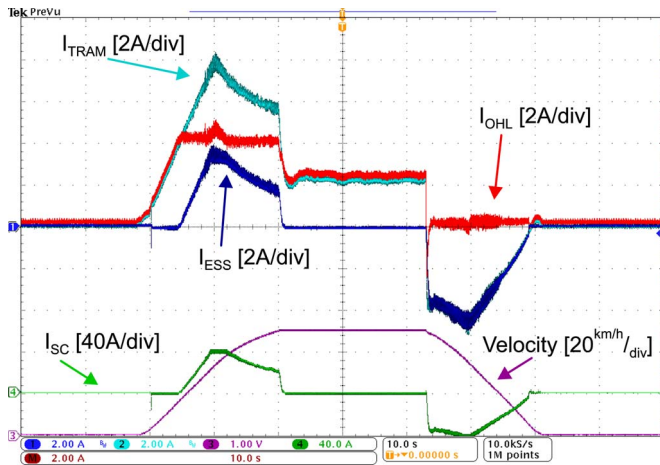


Fig. 24. Measurement results of the peak shaving PM: The strategy shaves I_{OHL} peaks (over 4 A) until I_{SC} is not limited.

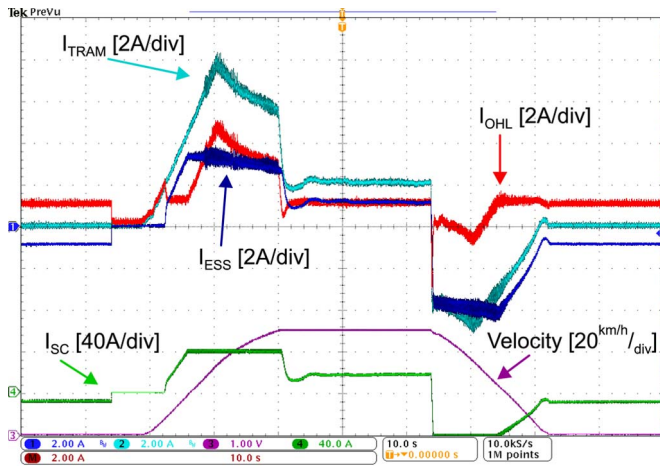


Fig. 25. Measurement results of the PM based on mean power: This PM tries to consume constant power that is equal to the mean tram power. The green arrow shows the moment of the charge break due to the full SC capacity.

cities, and ESSs have a major role in the tram efficiency. The recuperation rate of the Pilsen tram line is very high; it is approximately 25%.

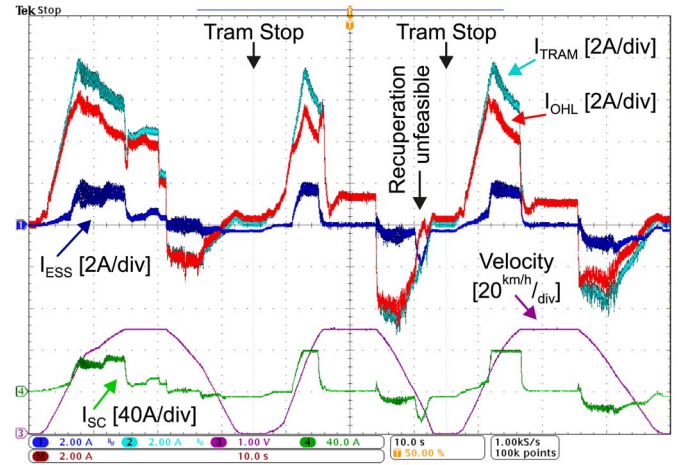


Fig. 26. Measurement results of the T-S fuzzy PM: The different behavior compared with basic strategies is mainly visible during braking. The recuperation is preferable.

The energy consumption savings are qualitative criteria of the PM comparison. These savings also include calculated OHL losses. Results of this PM are shown in Fig. 27.

A. Proportional PM

The proportional PM can reach energy savings up to 20% in case of the disabled recuperation and 1.5% in case of common line recuperation (first column in Fig. 27). This strategy covers the proportional part of the power demand (depending on k_{PROP}) or power supply. It decreases the recuperated energy and increases ESS losses. Therefore, the energy storage losses can reach 4%, which is comparable to braking resistor losses in case of the tram without an ESS.

B. Peak Shaving PM

Results of the peak shaving PM for different peak levels are shown in the second column in Fig. 27. This PM allows reaching similar energy savings such as proportional PM in cases of disabled recuperation. Energy savings in the case of common line recuperation are higher than savings with using the proportional PM. However, they are still too low due to the decreasing recuperation again. This strategy is useful in locations with long power supply distances or high OHL resistance (worn-out wire) to decrease the OHL losses.

C. PM Based on Mean Power

Results of this strategy are shown for different k_{MEAN} parameter in the third column in Fig. 27. The disadvantage is visible from graphs of both recuperation cases (enabled/disabled). The best results are reached for k_{MEAN} equals zero. However, it is the same strategy as proportional with k_{PROP} equals 1.

The weakness of the strategy is in charging of the energy storage before braking (due to positive mean power). Therefore, the ESS is not prepared (discharged) to store the kinetic energy. The braking energy must be recuperated to the OHL or wasted in the braking resistor (if the recuperation is not feasible). This

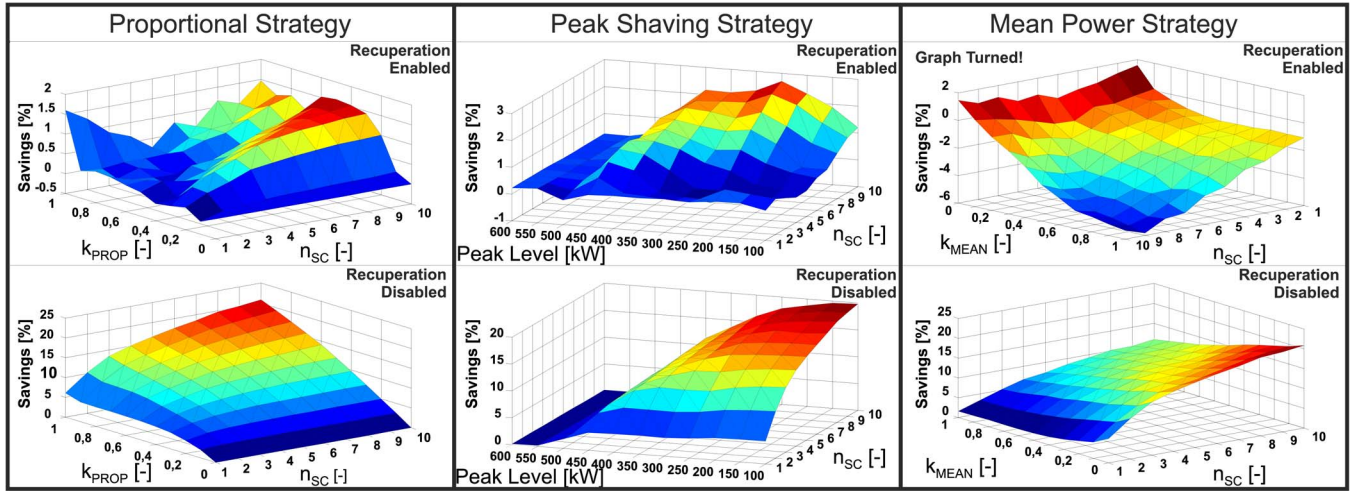


Fig. 27. Proportional PM results (savings depend on the number of supercapacitors n_{SC} and proportional constant k_{PROP}) compared with the tram without an ESS. Peak shaving PM results (savings depend on the number of supercapacitors n_{SC} and peak level) compared with the tram without an ESS. Mean power PM results (savings depend on the number of supercapacitors n_{SC} and multiplying constant k_{MEAN}) compared with the tram without an ESS.

TABLE II
PM ENERGY SAVINGS

Energy Savings - 3 SC modules		
Recuperation to OHL	Enabled	Disabled
Proportional	0,4%	12,5%
Peak Shaving	1,0%	6,3%
Mean Power	-0,6%	10,4%
Mean Power Predicted*	-0,5%	9,4%
T-S Fuzzy Default	6,9%	14,2%
T-S Fuzzy after Dif. Evolution	7,5%	14,9%

strategy uses the energy storage very often. Thereby, losses in the ESS are excessively high.

D. Comparison of T-S Fuzzy and Basic PM

Table II compares results of basic PM strategies against a novel T-S fuzzy. The mean power predicted PM method [24] is derived from the former mean power PM approach. Mean power predicted PM is based on the knowledge of the tram line consumption profile defined for selected time horizon, which varied in our tests from 20 to 220 s (the best results used for energy savings comparison were achieved for the horizon of 120 s).

Both simulated T-S fuzzy PMs have significantly lower losses than the basic methods. The T-S fuzzy optimized by DE for common tram line recuperation and the three SC 125 V/63 F modules have the best results. It should be remarked that T-S fuzzy was optimized for three SC modules—If we change the number of SC modules, T-S fuzzy should be retuned.

E. Return on the Investment

Estimation of the return on the investment depends on many factors such as electrical energy costs, initial investment (supercapacitor, dc/dc converter, and system installation), and tram duty cycle.

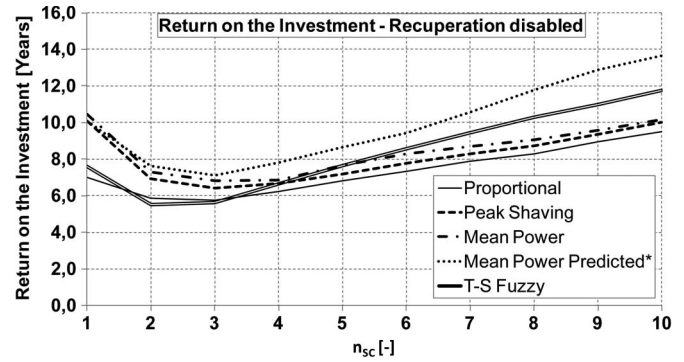


Fig. 28. Return on the investment—recuperation disabled.

Presumed parameters:

- Real tram line no. 4 in Pilsen, Czech Republic, 7 km long.
- Electrical energy costs 74 euro/MWh— $COST_E$.
- Supercapacitor Module (125 V/63 F) 5000 euro— $PRICE_{SC}$.
- dc/dc converter incl. installation 7400 euro— $PRICE_{CON}$.

$$PRICE_{Installation} = N_{SC} \cdot PRICE_{SC} + PRICE_{CON}$$

$$MONEY_{\frac{Savings}{Year}} = ENERGY_{\frac{Savings}{Drive}} \cdot COST_E \cdot \frac{DRIVES}{YEAR}$$

$$ROI = \frac{PRICE_{Installation}}{MONEY_{\frac{Savings}{Year}}}$$

The following graphs (see Figs. 28 and 29) show return on the investment in years depending on the number of SC modules.

This economic analysis shows that a number higher than four supercapacitor modules is not profitable. The lifetime of the SC is ten years for permanent maximal SC voltage or 1 000 000 full charging cycles. The T-S fuzzy PM uses SC gently, and expected lifetime could be up to 20 years. After the lifetime expiration, the SC is still usable but with 20% reduced capacity. It is evident that only the T-S fuzzy PM is suitable for tram lines with a high recuperation rate.

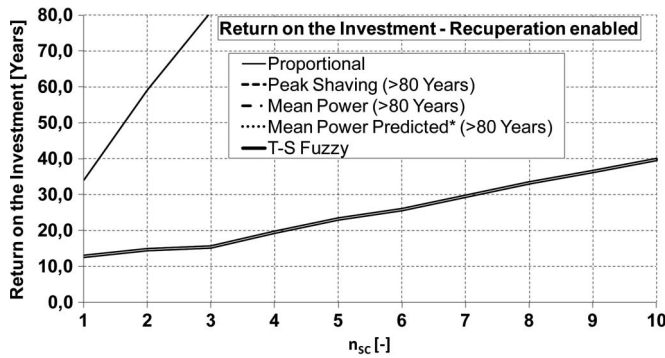


Fig. 29. Return on the investment—recuperation enabled.

IX. CONCLUSION

This paper has presented essential results of our research on position-based T-S fuzzy PM for a human-driven tram with an ESS.

The introduced novel T-S fuzzy PM achieved significantly better results than the commonly used PM. Due to the known tram location and previous scenarios, the PM can be prepared for future energy demands. The presented soft computing technique leads to dramatic reduction of problem dimensionality. Moreover, this approach uses two prediction horizons. Long-term energy prediction allows estimation of both position and size of energy peaks on the long distance ahead the tram, whereas short-term energy prediction provides the controller with mean energy estimation demanded for the actual track section. These features bring improved accuracy of our method in comparison with the existing PM strategies.

The T-S fuzzy system was tuned offline by a heuristic optimization algorithm (DE). The real-time part of the algorithm was implemented in common fixed-point DSP TMS320F2812 and, therefore, is ready for easy installation into a vehicle.

The calculated return on the investment for the simulated real tram line shows that competitive control strategies of the onboard ESS will be useful only with disabled recuperation to the OHL. On the contrary, the presented T-S fuzzy PM significantly reduces the payback period under all operating conditions. In order to achieve the best control performance, the PM strategy must be exactly tuned for the given number of supercapacitor modules.

The T-S fuzzy PM algorithm is independent on the ESS type and can be simply adapted for other ESS technologies.

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